

Compute for every scale

ISC Hamburg, June 22, 2026

Moritz Thüning

CONFIDENTIAL - CONTAINS TRADE SECRETS



Tenstorrent at a Glance



Founded
March 2016



Capital Raised
\$1B+

Investors



Target Markets



Data Center



Automotive / Robotics



Edge AI / IoT

\$400B

AI / Compute TAM

Key Performance Indicators

~1,000 Employees

700+ R&D Professionals

900+ Developers on TT platform

Global Footprint



Business Model

- Tenstorrent monetizes its AI and RISC-V Technology in 3 ways:
- Licenses AI and RISC-V IP to customers building silicon
 - Sells AI Chiplets, RISC-V Chiplets, & AI Chips
 - Sells AI boards, servers and systems

Compute for every scale.



Cards

Add to, or config your own rig. Passive, active and liquid cooling options.

Starting at \$999

[Buy Blackhole®](#)



Workstations

Whisper-quiet, liquid cooled - run models up to 120b parameters from your desk.

Starting at \$9,999

[Buy TT-QuietBox®](#)



Servers

Sovereign, scale-out servers for production AI.

Starting at \$70,000

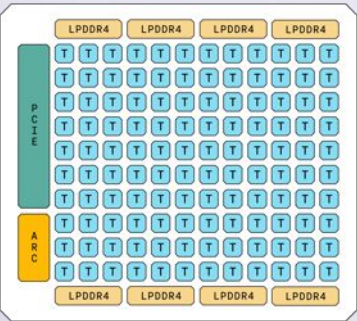
[Explore Tenstorrent Galaxy™](#)

Tenstorrent Hardware and IP

Devices

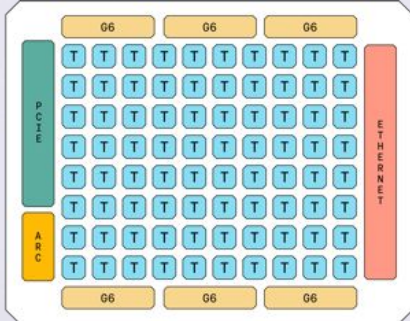
Grayskull®

AI Processor



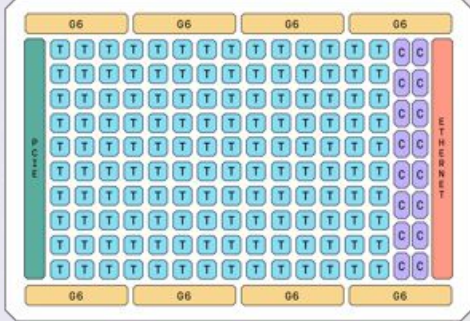
Wormhole™

Networked AI Processor



Blackhole™

Standalone AI Computer



AI Processors

IP available
for licensing

Tensix
NEO AI IP



PCIe Cards



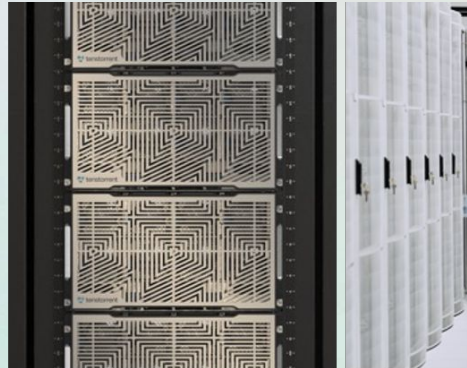
TT-QuietBox™



TT-LoudBox™



Tenstorrent Galaxy™



RISC-V
Ascalon
CPU IP



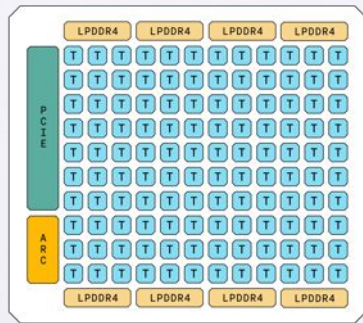
Core Silicon Roadmap

2023

High Perf AI ASIC

Grayskull®

AI Processor



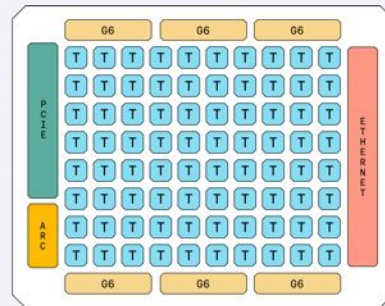
- 120 Tensix Cores
- 12nm
- 332 TFLOPS (FP8)
- 83 TFLOPS (BLOCKFP8)
- 16 lanes of PCIe Gen 4
- 8 channels LPDDR4

2024

Scalability

Wormhole™

Networked AI Processor



- 80 Tensix+ Cores
- 12nm
- 262 TFLOPS (FP8)
- 148 TFLOPS (BLOCKFP8)
- 16 lanes of PCIe Gen 4
- 16x100 Gbps Ethernet
- 6 channels GDDR6

2025

Heterogeny

Blackhole™

Standalone AI Computer



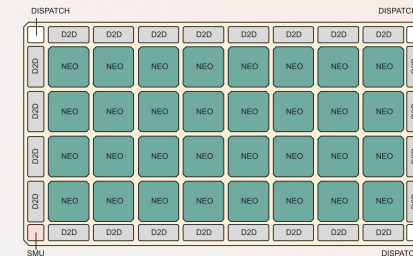
- 120 Tensix++ Cores
- 6nm
- 664 TFLOPS (FP8)
- 332 TFLOPS (BLOCKFP8)
- 10x400 Gbps Ethernet
- 8 channels of GDDR6
- 16 RISC-V CPU cores

2026-2027

Chiplets

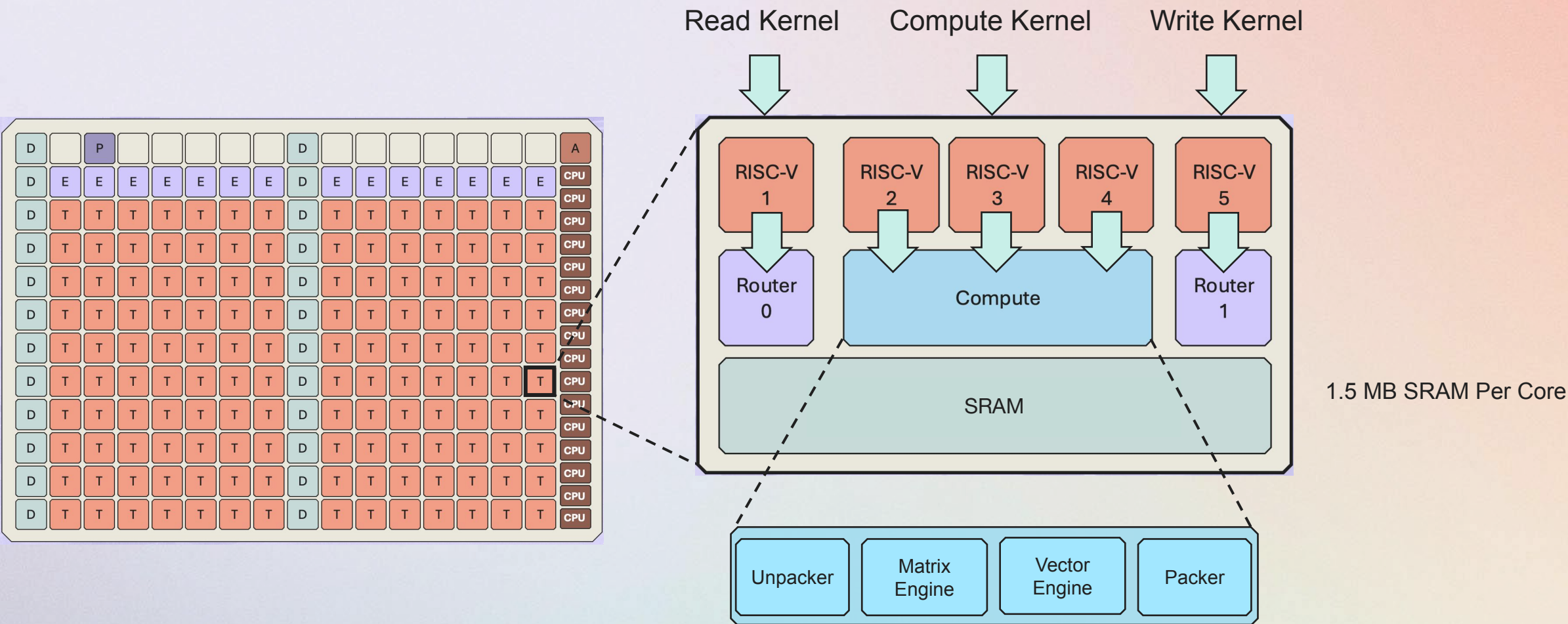
Quasar

Low Power AI Chiplet



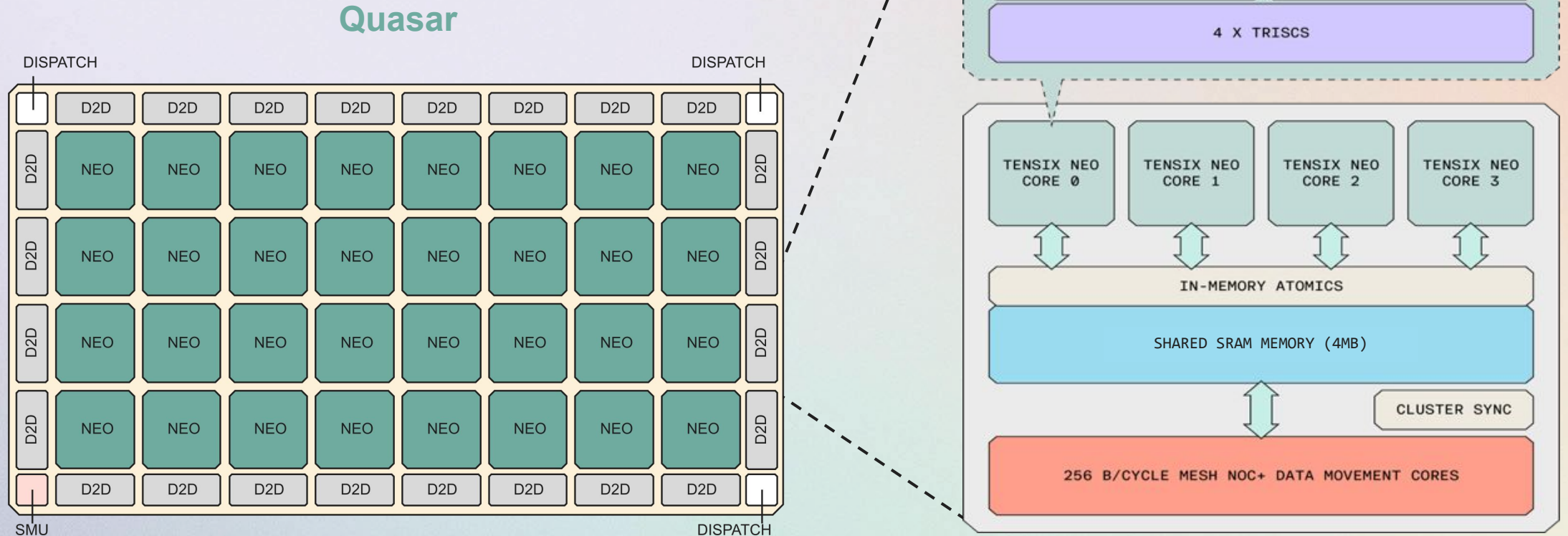
- 128 Tensix NEO Cores
- 4nm Chiplet
- Non-blocking D2D interfaces
- Easily stack Quasar or combine to choose your own compute

Blackhole Architecture



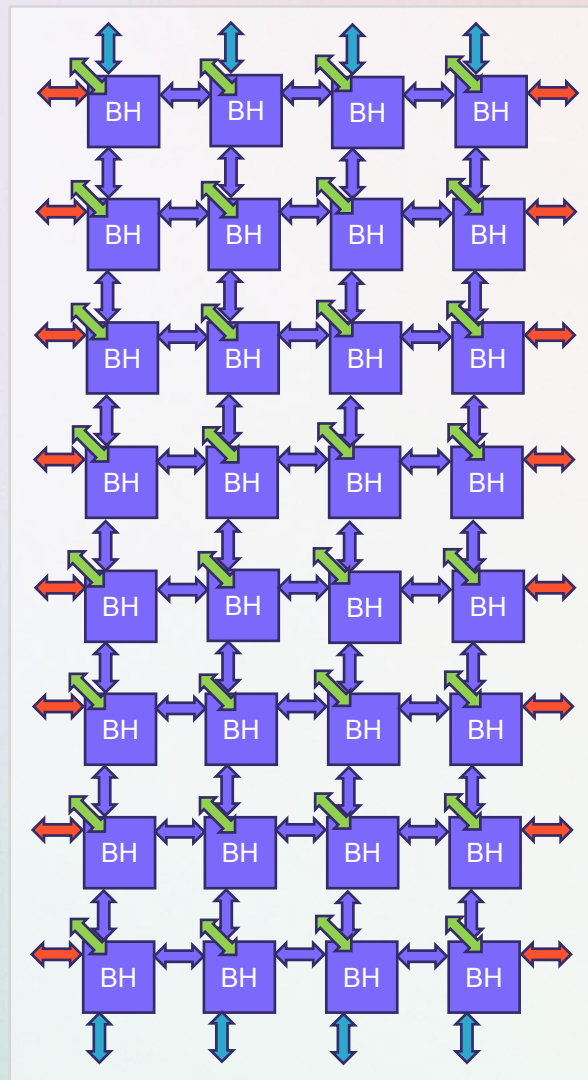
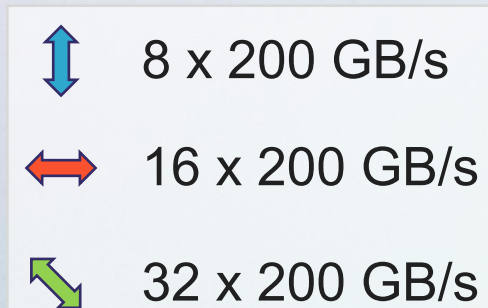
Beyond Blackhole: Tensix NEO IP and Quasar Chiplet

- 32 Tensix NEO Clusters
- 128 Tensix NEO Cores
- Adds support for MX precision formats



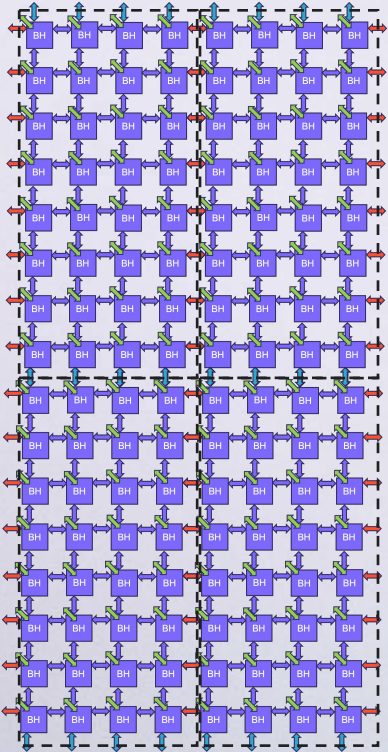
Tenstorrent Galaxy™ Blackhole Server: 32 Chips in a 4x8 Mesh

- GDDR6: 1 TB @ 16 TB/s
- Virtualize multiple Galaxies into exaflop-scale systems
- 11.2 TB/s Galaxy I/O

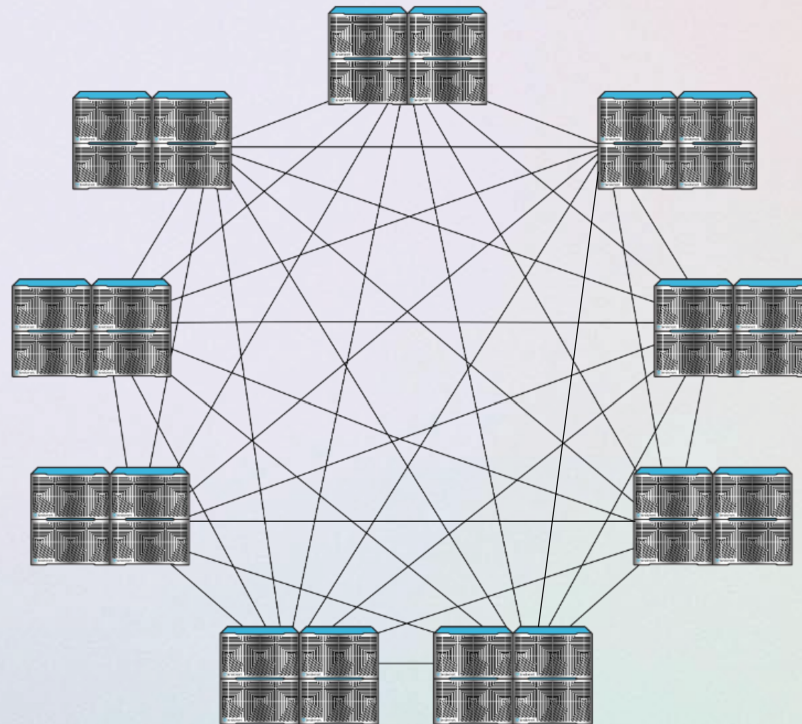


Flexible Scale-Out Options

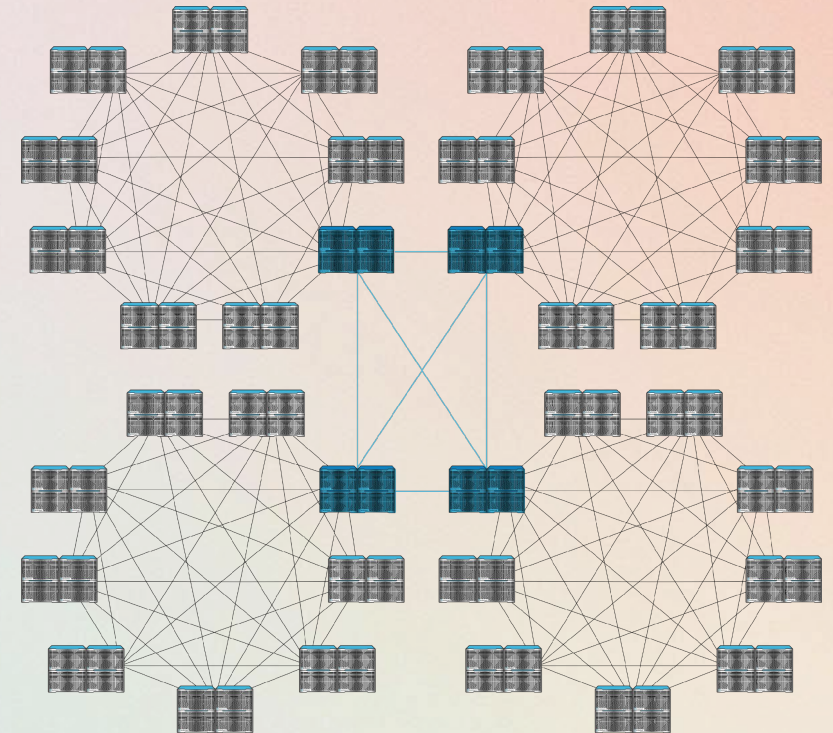
Quad



Tenstorrent Galaxy
Supercluster 36



Tenstorrent Galaxy
Supercluster 144



No switches required ➡ Only Ethernet cables!

Tenstorrent Open-Source Software Stack

TT-Forge: MLIR based compiler

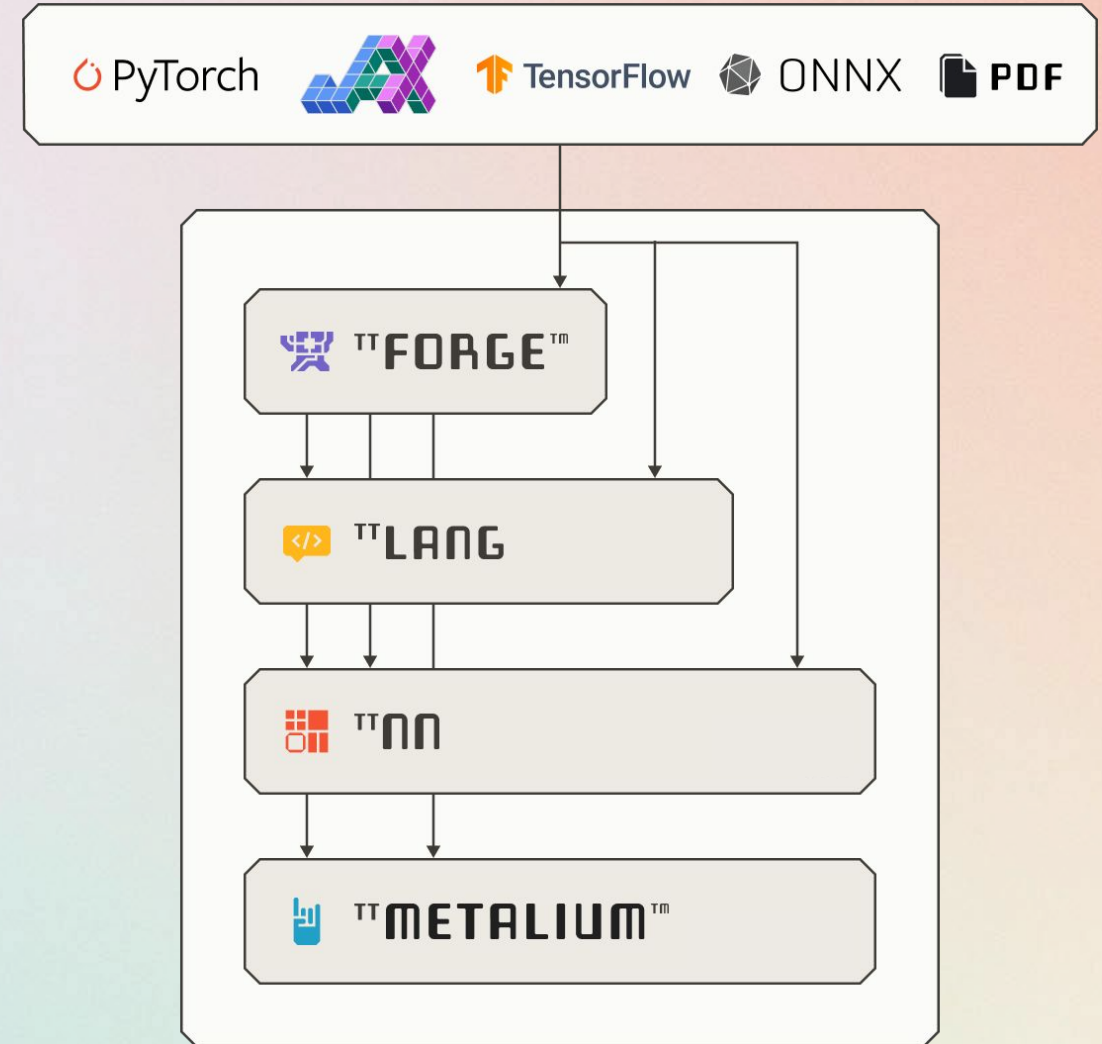
- Runs thousands of models
- <https://github.com/tenstorrent/tt-forge-models>

TT-Lang: Python DSL kernel language

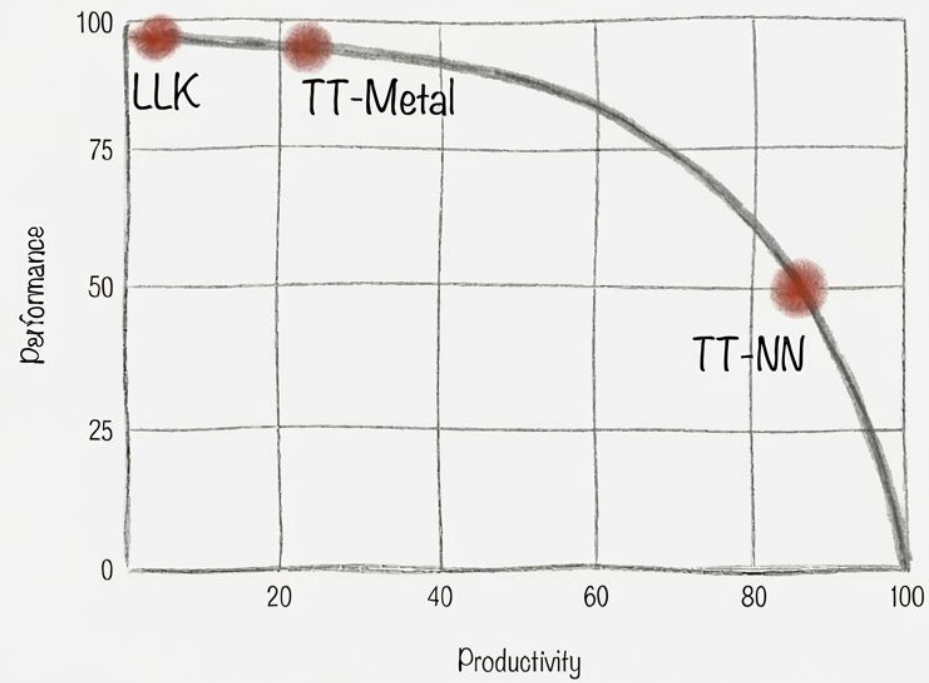
- For human-driven and AI-assisted code generation

TT-NN: Library of high-performance operators

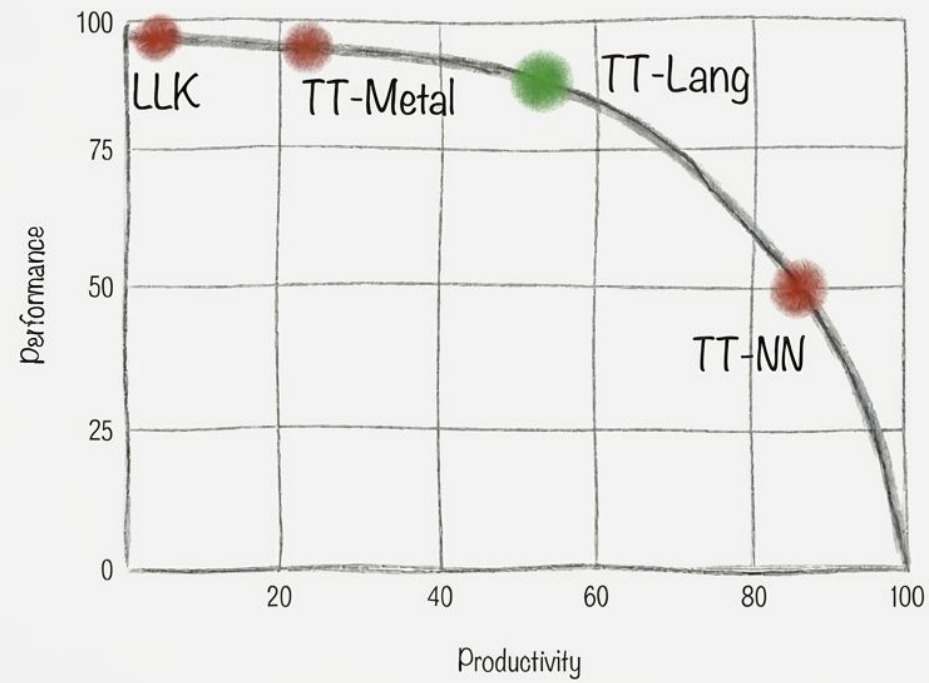
TT-Metalium™: Low-level C++ programming model for kernel development



Levels of Software Abstraction



TT-Lang targets the Sweetspot



Blazing Fast DeepSeek V3.2 671B

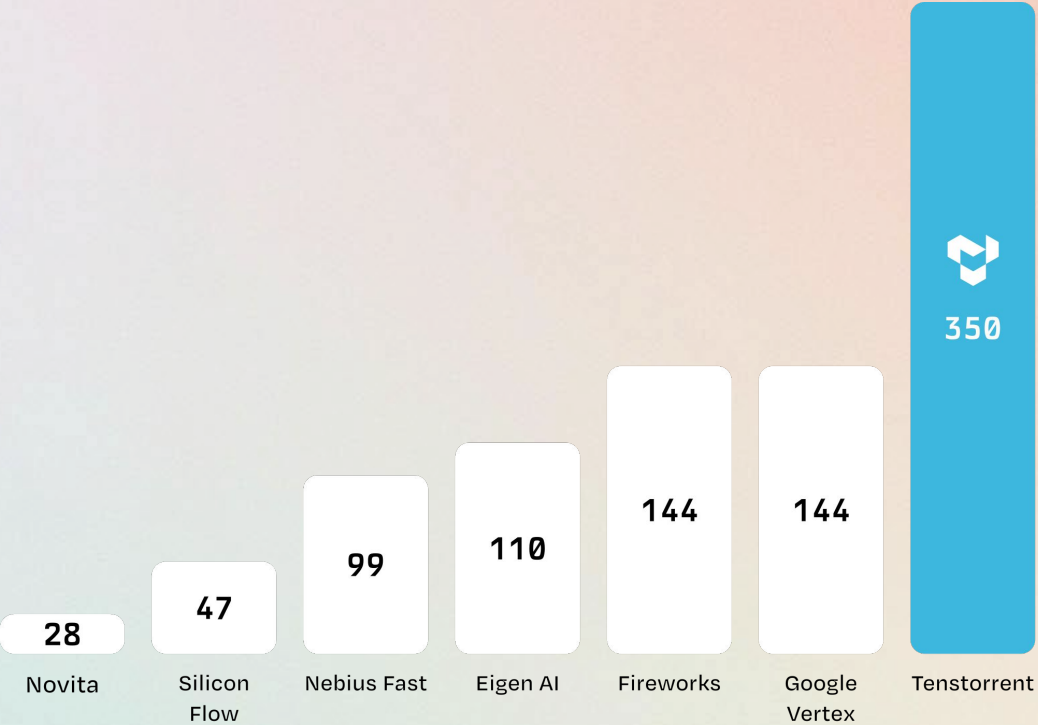
Prefill

Sequence Length: 100k



Seconds

Decode



Tokens / Second / User

Source: Artificial Analysis, top providers

Faster Than Real-Time Video Generation

- Wan 2.2 A14B:
81 frames of 720p in 2.4 sec



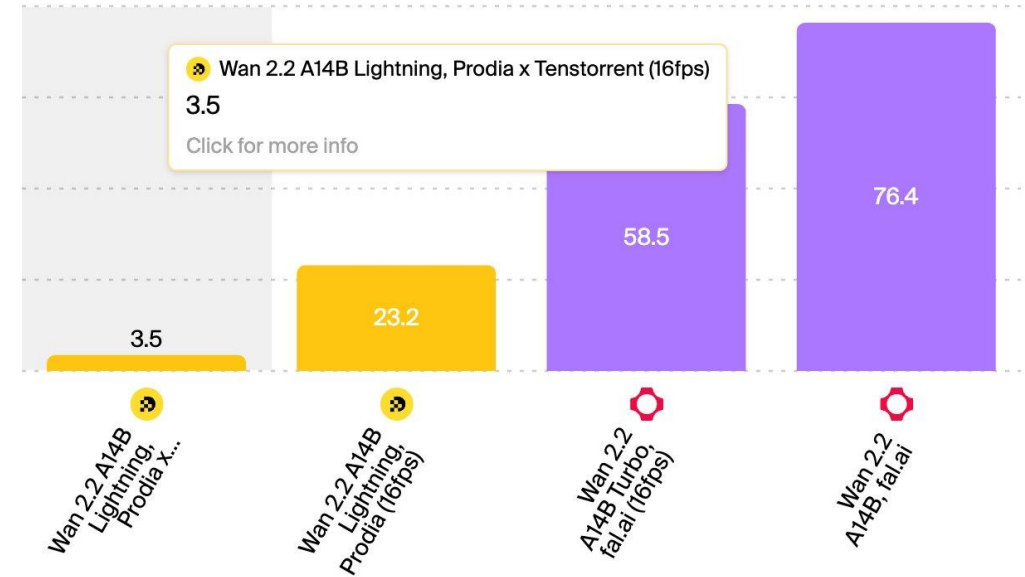
API Generation Time

Generation time: Seconds to generate a 1080p, 5s, 24fps video. When this configuration is not available for the model, the closest settings are used, Lower is better

4 of 4 providers



Artificial Analysis



Source: Artificial Analysis

Software Highlights

Inference & Serving *model deployment, from engine to GUI*

- **vLLM (Tenstorrent fork)** — the serving engine: runs the model and exposes an OpenAI-compatible API for high-throughput inference
- **tt-inference-server** — the orchestration layer around vLLM: manages containers, model downloads & serving config; authoritative source for validated models
- **TT-Studio** — the web GUI on top of tt-inference-server: point-and-click model deployment, no command line

Simulation & Emulation *develop without physical hardware*

- **ttsim** — full-system simulator of the Tenstorrent hardware itself (models the device/architecture)
- **tt-emule** — C++ emulator of the device-level kernel & host APIs (run TT-Metalium code on CPU, no chip needed)

Linux on Blackhole

- **tt-bh-linux** — boots a real Linux kernel directly on the Blackhole P100/P150 RISC-V cores



TT-Bio – Drug Design

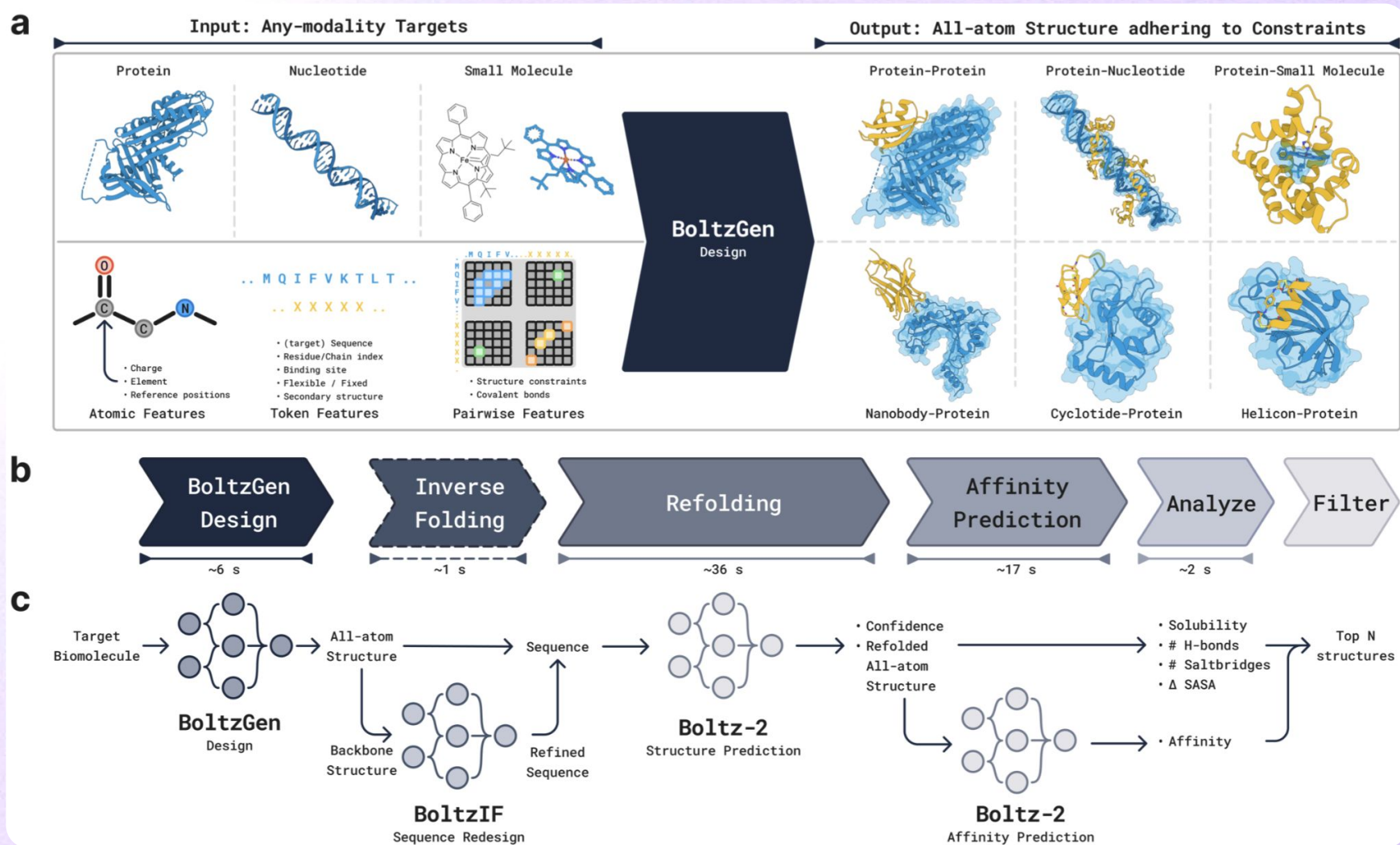
Boltz-2, BoltzGen, ESMFold-2, Protenix-v2 on Tenstorrent



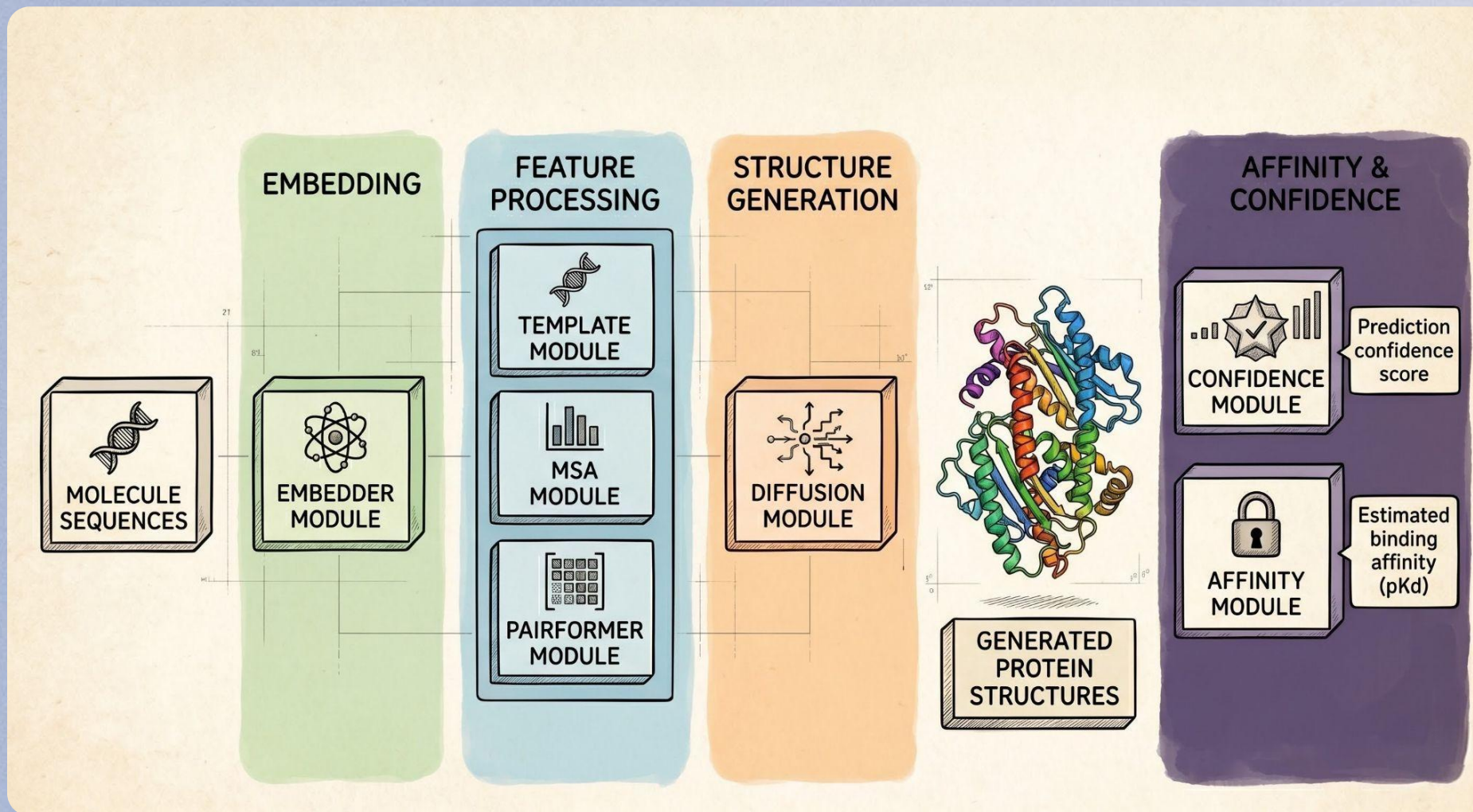
Biomolecular Structure Prediction (Boltz-2 $\approx$ AlphaFold 3)

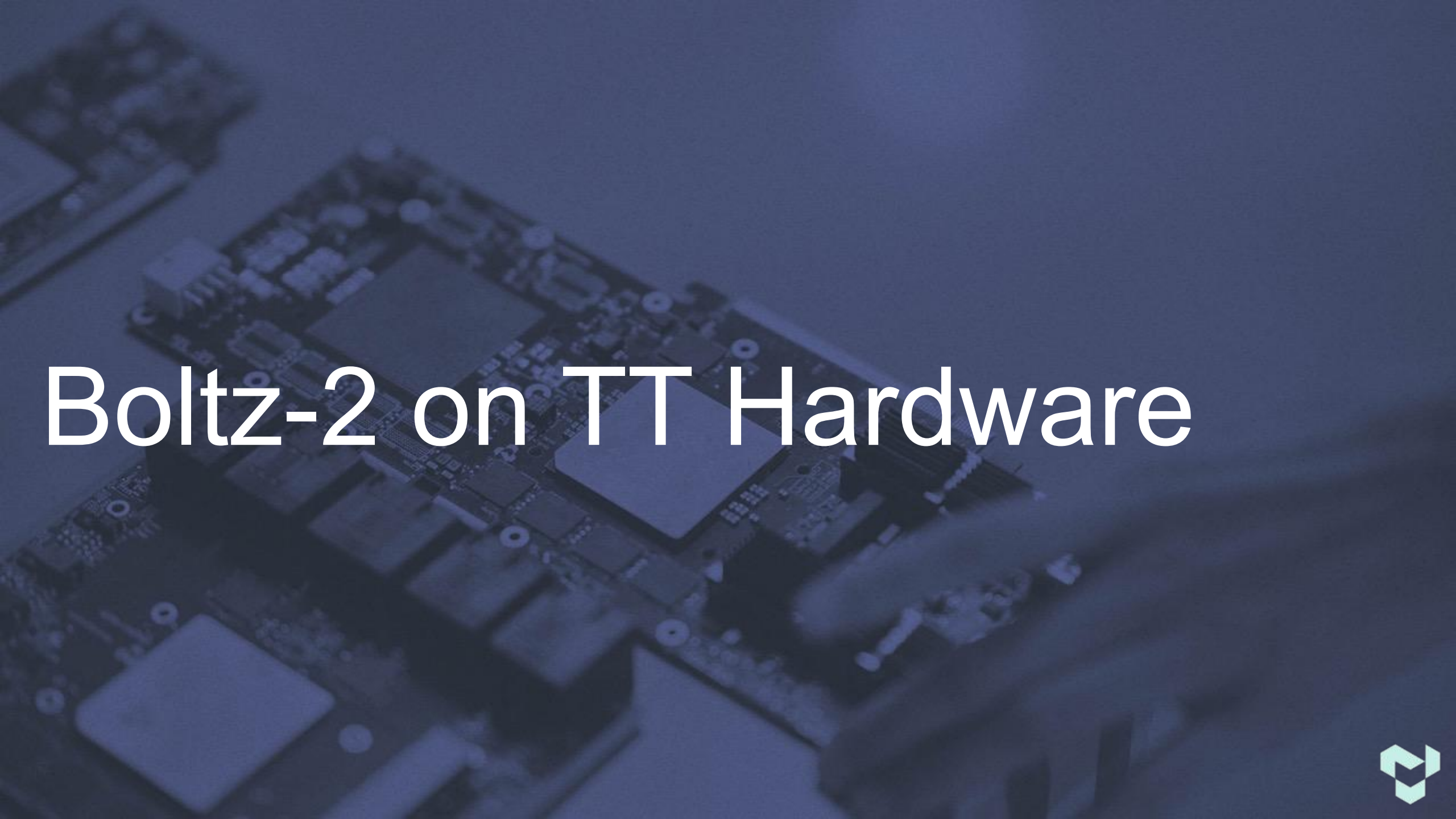


BoltzGen: State-of-the-art Drug Design Pipeline



Boltz-2 Architecture





Boltz-2 on TT Hardware



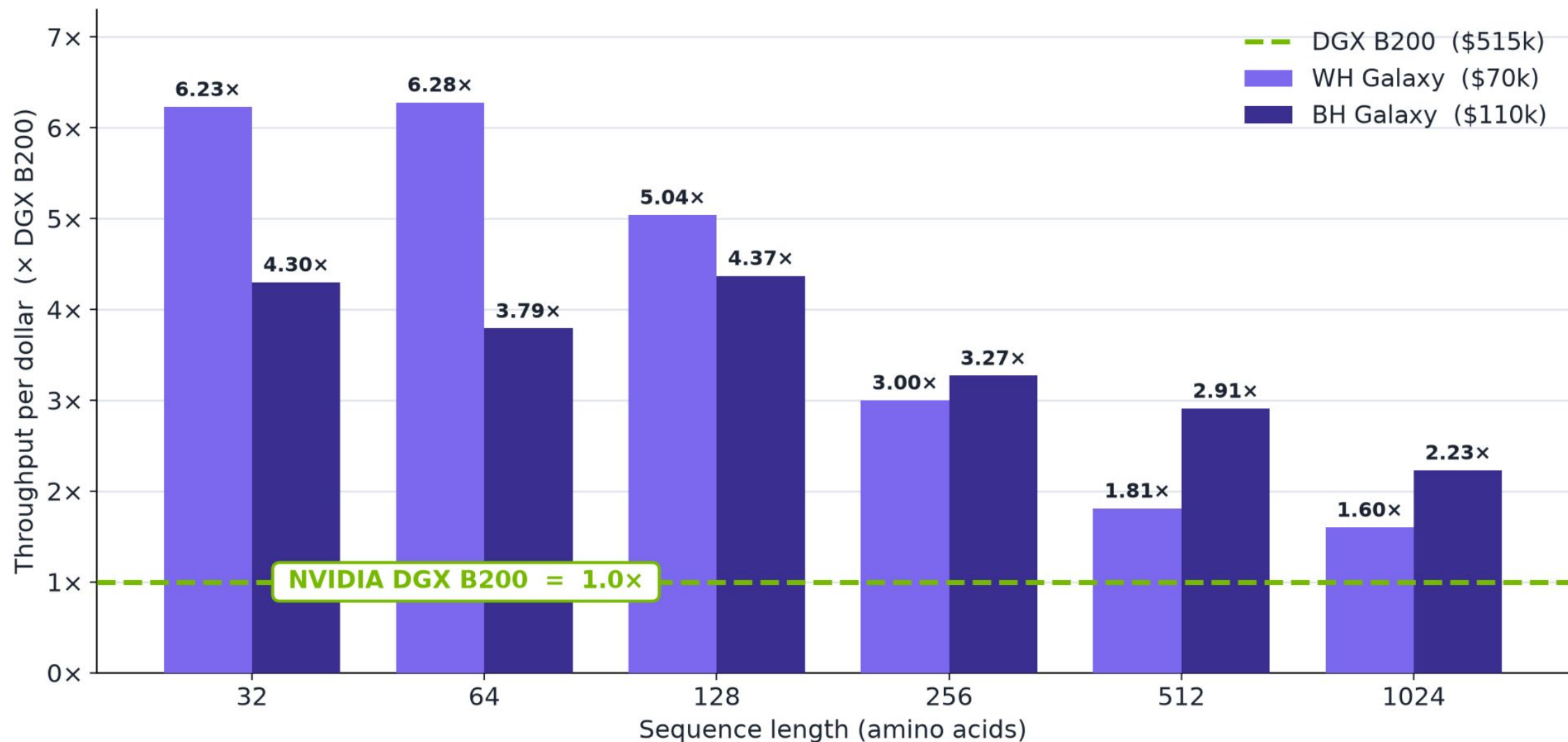
Boltz-2 on BH Galaxy

- Boltz-2 **parameters** are only **2 GB**
- **Data Parallelism**
Replicate model, split data
- **Linear Scaling**
BH Galaxy has **32x** the throughput of a single BH processor
- **Throughput** more important than latency

tt-boltz 32 devices 0/64 (0%) 00:00:19				
device 0	·		Loading model...	0/2
device 1	·		Loading model...	0/2
device 2	·		Loading model...	0/2
device 3	·		Loading model...	0/2
device 4	·		Loading model...	0/2
device 5	·		Loading model...	0/2
device 6	·		Loading model...	0/2
device 7	·		Loading model...	0/2
device 8	·		Loading model...	0/2
device 9	·		Loading model...	0/2
device 10	·		Loading model...	0/2
device 11	·		Loading model...	0/2
device 12	·		Loading model...	0/2
device 13	·		Loading model...	0/2
device 14	·		Loading model...	0/2
device 15	·		Loading model...	0/2
device 16	·		Loading model...	0/2
device 17	·		Loading model...	0/2
device 18	·		Loading model...	0/2
device 19	·		Loading model...	0/2
device 20	·		Loading model...	0/2
device 21	·		Loading model...	0/2
device 22	·		Loading model...	0/2
device 23	·		Loading model...	0/2
device 24	·		Loading model...	0/2
device 25	·		Loading model...	0/2
device 26	·		Loading model...	0/2
device 27	·		Loading model...	0/2
device 28	·		Loading model...	0/2
device 29	·		Loading model...	0/2
device 30	·		Loading model...	0/2
device 31	·		Loading model...	0/2

Boltz-2 on Tenstorrent: up to 6x the throughput of similar priced GPU cluster

Protenix v2: similar throughput per \$



Get started

- Try it out
 - <https://console.tenstorrent.com/>
- Contribute
 - <https://github.com/tenstorrent>
- Join our Discord
 - <https://discord.gg/tenstorrent>



Try the fastest inference

 DeepSeek R1 685B

Industry-leading token speed

Open Chat →

LIVE

Generate a video in seconds

 Wan 2.2

8x faster at half the cost

Generate Video →

LIVE

PLATFORM

Agents

Ready-to-deploy AI solutions. Agents, bots, coding assistants — powered by the stack below. Connect to Slack, APIs, or your IDE.

 Deploy a bot →
Your Claw in 1 click

 Code agent →
VS Code, Cursor, Continue

Models

AI as an API. Chat completions, image, video, audio generation, fine-tuning — OpenAI-compatible. Change your base URL and go.

 Get API Key →
OpenAI-compatible

 Chat →
Fastest inference

 Video →
8x faster generation

 Image →
Text to image

 Audio →
Speech & transcription

 Train →
Finetune on your data

Deployments

Your own model endpoints. Pick a model, deploy it for your org. Pay per hour, not per token.

 Deploy model →
Dedicated endpoint

 Browse catalog →
Available models

Compute

On-demand servers with Tenstorrent accelerators. Full shell access, VSCode in the browser. Run your own code.

 Baremetal →
SSH access

 Dev environment →
Managed instance

Hardware

Buy Tenstorrent cards and Galaxy systems. Run your own cluster, on-prem or in your cloud.

 Explore systems →
Wormhole, Blackhole, Galaxy



Thank you!

